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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Regular / Supplementary Examinations Feb-2024

ELEMENTS OF MECHANICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.
3. Steam table book is permitted.

PART-A

1	1.1	Polyester is an example of _____ type of polymer.	01
	1.2	Define bore of an IC engine.	01
	1.3	Connecting rod of IC engine is made of _____ material.	01
	1.4	Mention the types of flames in oxyacetylene welding.	02
	1.5	Filler material commonly used in soldering process is known as _____.	01
	1.6	_____ type of turbine is mixed flow turbine.	01
	1.7	Refrigerator works on _____ law of thermodynamics.	01
	1.8	Magnetic particle inspection is carried out for _____ type of material.	01
	1.9	_____ is the equation used to find out the taper angle to obtain taper shape by using swiveling of compound rest method.	02
	1.10	In _____ type of control system, the feedback system is not present.	01
	1.11	The process of reducing diameter of cylindrical workpiece in lathe is called _____ operation.	01
	1.12	Define robot.	01
	1.13	Classify the types of robot based on its configuration.	02
	1.14	In _____ type of steam turbine, the pressure reduces suddenly before entering the turbine.	01
	1.15	Constant volume cycle engines are also called as _____ type of engines.	01
	1.16	Which are the gears that are used to transmit the motion when the shafts are parallel to each other?	02

PART-B

2	a	Differentiate between Thermosetting and thermo plastic polymers.	06
	b	What are composite materials? Classify them & explain the applications of different types of composite materials.	10
3	a	With a neat sketch, briefly explain how to obtain taper turning in lathe machine by using swiveling of compound rest method.	06
	b	How do you carry out the electric arc welding process? Explain with a neat sketch & discuss the general defects in welding products.	10

		OR	
4	a	When do you recommend NDT application during the life cycle of the component? Briefly explain the magnetic particle inspection process with a neat diagram.	10
	b	With a neat diagram explain the working of oxy-acetele welding process.	06
5	a	With a neat sketch explain the working of high head water turbine.	08
	b	Explain with a neat sketch, the Temperature-Enthalpy diagram for formation of steam at constant pressure.	08
		OR	
6	a	With a neat sketch, explain the working of vapour compression refrigerator.	08
	b	With the help of $P-V$ and $T-S$ diagram, explain the working of Brayton cycle for closed cycle gas turbine.	08
7	a	Explain with schematic diagram working principle of IC engine in which burning of fuel takes place at constant volume.	08
	b	The following observations were made during a trial on a $4-S$ diesel engine: $Cylinder\ dia\ = 25cm$ | $Mean\ effective\ pressure = 6\ bar$ $Stroke\ of\ piston\ = 40cm$ | $Diesel\ oil\ consumption = 0.1m^3/min$ $Crank\ shaft\ speed\ = 250rpm$ $Brake\ load = 70kg$ $Brake\ drum\ dia = 2m$ $Specific\ gravity\ of\ diesel\ = 0.78$ $Calorific\ value\ of\ the\ diesel = 43900kJ/kg$ Find $BP, IP, FP, \eta_{mech}, \eta_{\beta\ thermal}, \eta_{I\ thermal}$	08
		OR	
8	a	With a neat sketch explain series-parallel type of hybrid electric vehicles.	08
	b	With a neat sketch derive the velocity ratio and Train value for simple and compound gear train.	08
9	a	Enumerate mechatronics control system using engine management system as an example with an appropriate diagram.	10
	b	Differentiate between traditional and mechatronics approach of control system.	06
		OR	
10	a	With a neat sketch, explain the anatomy of Robot.	08
	b	Explain with a flow chart the phases of mechatronic system design process.	08